# Microbiology

## general :-

* SN ① Koch's postulates and Koch's phenomenon
* ★ ② difference prokaryotic and eukaryotic
* ★ ③ difference b/w gram ⊕ and gram ⊖ cell wall
* SN ④ flagella
* SN ⑤ fimbriae
* SN ⑥ Bacterial spore
* SN ⑦ Bacterial growth curve
* SN ⑧ Methods of sterilisation
  - (i) Hot air oven
  - (ii) Auto clave
  - (iii) filtration
  - (iv) Radiation
  - (v) Aldehydes
* ⑨ culture media → Practical
* SN ⑩ Mutation
* LD ⑪ gene transfer
  - ★★ (i) Transformation
  - ★★ (ii) transduction
  - (iii) lysogenic conversion
  - ★★ (iv) conjugation
* ★ ⑫ PCR
* ★★★ ⑬ difference between Exotoxin & Endo toxins.

## Immunology :-

⑭ Immunity :-

defination, types
difference between Active & passive immunity.

⑮ Antigen :-

defination, types, factors affecting
SN Super antigens.
Antigenicity.

SN ⑯ Immunoglobulin IgG, IgA, IgM, IgE

LQ (17) Precipitation Rxn :-

Prozone phenomenon
Mechanism
Application, types

LQ (18) Agglutination Rxn :- define, types.

R (19) CFT

R (20) Immunofluorescence

★★★★★ (21) ELISA

(22) Compliment pathway :- classical, Alternative ] only cycle

* Biological effect of compliment

★★ (23) difference between T cells & B cells.

(24) MHC and HLA typing.

★★ (25) diff. between primary and secondary Humoral responses

★★★★★ (26) Monoclonal Antibody.

★★★★★ (27) Hypersensitivity Rxn

(28) * difference immidiate and delayed type.

SN → (i) Atopy (ii) Pk Rxn (iii) Arthus Rxn (iv) Type IV (v) swartzman Rxn

(28) Allograft Rxn and Prevention of graft Rejection. transplantation Types.

SN (29) graft v/s Host Reaction.

SN (30) Immunosurveillance

(31) complication of Blood transfusion.

Bacteriology :-

★★ (32) Staphylococcus Aureus → Morphology, Pathogenesis, lab diagnosis

SN → MRSA

★ (33) Streptococcus → Morphology, Pathogenesis, lab diagnosis

SN ⇒ CAMP test

★★★ (34) Pneumococcus → Morphology
Pathogenesis
lab diagnosis

★ difference between str. Pneumoniae & str. viridance

★★★ (35) N. Meningitidis
&
N. gonorrhoeae → Morphology
Path. genesis
lab diagnosis
Culture

★★ difference between N. meningitidis and N. gonorrhoeae.

(36) Corynebacterium diptheriae → Morphology
Culture
Pathogenesis
lab diagnosis
Pyz test, Eleks gel ppt test, Schick test

★★★ (37) Bacillus → Morphology
Culture
Biochemical Reaction
Pathogenesis
lab diagnosis

(38) Clostridium tetani → Morphology
Pathogenesis
lab diagnosis
toxins

SN → (i) Nagles Rxn.
(ii) Reverse CAMP test.
(iii) Botulinums.

★★★ LQ (39) E.coli → Morphology
Culture
toxins
Pathogenesis
lab diagnosis

** LO (40) Shigella → Morphology, culture
Pathogenesis, lab diagnosis
toxins, classification

** LO (41) Salmonella → Morphology, culture
Pathogenesis, lab diagnosis
Antigenic structure

** Widal test

(42) Vibrio → Morphology
Pathogenesis, lab diagnosis
toxins

** difference between classical and ELTOR

SN (43) H. Pylori

(44) Pseudomonas → Morphology, culture
Pathogenesis, lab diagnosis
pigment

(45) Haemophilus → (i) X and V factor
(ii) satellitism

(46) Bordotella → Morphology
Pathogenesis, lab diagnosis

(47) Brucella → culture
Pathogenesis, lab diagnosis

***** (48) M.T.B. → Morphology, culture
Biochemical Reaction
pathogenesis, lab degnosis

(i) diff b/w M.TB. and M.bovis
(ii) Tuberculin skin test
(iii) M.TB. and Non-tuberculous Microbacteria

(49) M. leprae → Morphology
Pathogenesis, lab diagnosis

(i) lepromin test

★★★★★ (50) Spirochaetes → T. Pallidum → Morphology
Pathogenesis, lab diagnosis

SN → VDRL test and RPR test

(51) Mycoplasma → characteristics culture
Pathogenesis, lab diagnosis

★★★ (52) chlamydia → Morphology, culture
Pathogenesis, lab diagnosis

SN → (i) LGV (2) Weil-Felix Reaction

## Virology

★ (53) Replication of viruses
★★★★★ (54) cultivation of viruses
★ (55) Interaction b/w Viruses
★★ (56) Inclusion bodies.
★ (57) Bacteriophage → morphology, life cycle
(58) Herpes virus → classification
Infection
Pathogenesis, lab diagnosis

SN → (i) cytomegalovirus (CMV)
(ii) EBV

(59) Picorna virus → classification,

(60) Polio virus → Pathogenesis, lab diagnosis
Prophylaxis

★★ → difference between OPV and IPV

(61) Influenza virus → Morphology
Pathogenesis, lab diagnosis

(*) difference between
***** (i) Orthomyxovirus & Paramyxovirus
***** (ii) Antigenic shift & Antigenic drift

(62) Arboviruses →
① Japanese Encephalitis virus.
② yellow fever
*** ③ dengue

** LO (63) RHabdo viruses → Rabies virus (LO) → Morphology
Pathogenesis, lab diagnosis
prophylaxis

** LO (64) Hepatitis - (B) virus →
Morphology, mode of transmission
clinical features, lab diagonis
prophylaxis

(**) difference b/w Hep. B & Hep. A Viruses.

Parasitology

** LO (65) Entamoeba histolica → Morphology, culture, life cycle
Pathogenesis, lab diagnosis
clinicle features

** diff. b/w Amoebic & basillary dysentry.

* (R) SN (66) giardia lamblia
* (R) SN (67) trichomonas vaginalis
***** LO (68) liesmania donovani → Morphology, culture, life cycle
Pathogenesis, lab diagnosis.
clinicle features.

SN (69) Chaga's disease
***** LO (70) plasmodium → life cycle, clinical features
Pathogenesis, lab diagnosis
***** complication of plasmodium falciparum.

★★★★★ (10) (71) Toxoplasma gondii → Morphology, life cycle Pathogenesis, lab diagnosis

★★ (72) diff. b/w T. solium & T. saginata

★ SN (73) Hydatid cyst and casoni test

★★★★★ LO (74) Ascaris lumbricoids → Morphology, life cycle Pathogenesis, lab diagnosis clinicle features

★★★★★ LO (75) Hook worms (Ancylostoma duodenale) → Morphology, life cycle Pathogenesis, lab diagnosis clinicle features

★ SN (76) NIH swab

R LR (77) W. Bancrofti → Morphology, life cycle Pathogenesis, lab diagnosis

Enumerate (78)

(78) oviparous nematodes.

(79) viviparous nematodes.

(80) ovo-viviparous nematodes.

★★ (81) Parasite infecting intestines.

(82) Parasite causing cutaneous and visceral larva migrans.

★★ (83) Bile stained egg.

★★ (84) eggs float in saturated salt solution.

★★ (85) eggs do not float in saturated salt solution.

(13) Retroviruses : HIV – Morphology, Mode of transmission, Pathe., Clinical features, Lab diagnosis

Opportunistic infection Associated with HIV

(15) Oncogenic viruses Enumerate

(16) Viruses Asso. with Human Cancer

## Mycology

(1) Classification of Fungi and Lab diagnosis

(2) Dermatophytes

(3) Mycetoma

(4) Histoplasmosis

(5) Cryptococcosis

(6) Candidiasis

(7) Aspergillosis

(8) Pneumocystis jiroveci

## clinical microbiology

(1) UTIs

(2) Meningitis

(3) STDs

(4) Stocks disc diffusion Method